

Nguyễn Đức Anh (Aldric)

Software Engineer · TypeScript · Python · Node.js

Product-minded engineer shipping end-to-end — from planning to production. Backend and full-stack across TypeScript, Python, and Node.js, with hands-on CI/CD, observability, and cloud delivery at startup pace.

Remote · anhnd350309@gmail.com · github.com/0xaldrich · linkedin.com/in/anhnd350309

Summary

Software engineer with 5+ years building and shipping production systems — from enterprise CRM at Teko to Web3 platforms supporting 50,000+ active users at 99.9% uptime. Strong product mindset, end-to-end ownership, and pragmatic judgment in fast-changing environments. Comfortable in modern AI-assisted engineering workflows, with deep experience across testing, CI/CD, observability, and cloud.

Experience

Senior Software Engineer · 5Solution

04/2025 — Present

Remote

Backend and full-stack delivery across a multi-product event-tech platform — 5sport, 5bib, 5ticket, 5pix — used by race organizers, athletes, and photographers.

- Lead end-to-end delivery for 5sport and 5bib: race registration, bib-to-photo matching, and live leaderboards — partnering with Product and Design from discovery to production.
- Designed the 5ticket event ticketing flow (QR check-in, anti-fraud, refunds) on Node.js/NestJS + PostgreSQL, hardened with idempotent APIs and integration tests.
- Built the 5pix high-throughput photo distribution pipeline — ingest, face/bib detection, CDN delivery — handling burst traffic on race days with autoscaling on Kubernetes.
- Hardened CI/CD with GitHub Actions, ephemeral preview environments, and structured logging + tracing for fast debugging and incident response.
- Adopted AI-assisted workflows (code review, scaffolding, test generation) — measurably faster cycle times without dropping coverage.

Stack — TypeScript, NestJS, Node.js, PostgreSQL, Redis, Kafka, Kubernetes, GitHub Actions, OpenTelemetry

Open-source Contributor · Mojave · 1sixtech

04/2025 — 07/2025

Remote · Open Source

Bitcoin / zk infrastructure — Rust workspace for full-node, sequencer, prover, and cryptographic utilities.

- Implemented block signing and verification (PR #17) — cryptographic identity guarantees for produced blocks.
- Built automatic ethrex upgrade tooling (PR #20) — keeping the workspace in lockstep with upstream changes.
- Worked across a multi-crate Rust workspace covering full node, sequencer, prover, and cryptographic primitives.

Stack — Rust, Cryptography, ZK, Bitcoin, ethrex

Senior Software Engineer · Topology Foundation

12/2024 — 03/2025

Remote

Decentralized Realtime Program (DRP) — open-source primitives for real-time collaborative apps.

- Improved build processes and CI/CD workflows for the DRP runtime, supporting modular code evolution.
- Authored high-coverage test suites — lifted code coverage to 90% across distributed components.
- Integrated performance metrics for network latency and CPU time, enabling real-time profiling and observability in decentralized environments.

Stack — TypeScript, Node.js, Vitest, GitHub Actions, Distributed Systems

Head of Engineering · AnyAxis Labs

02/2024 — 12/2024

Remote

Led the engineering org behind multiple Web3 products across the TON ecosystem.

- Led a high-availability backend supporting 50,000+ active users with 99.9% uptime.
- Built and directed a backend + blockchain team; owned hiring, technical roadmaps, and security posture.
- Rolled out microservices, DevOps practices, and CI/CD — standardizing deploys, observability, and incident response.
- Shipped TonPad (TON launchpad), AytuDex (50–100% faster execution than peer DEXes), and a Pump.fun-style platform with separated read/write/socket layers.

Stack — TypeScript, NestJS, Solidity, TON / FunC, PostgreSQL, Redis, Kafka, Kubernetes

Co-founder · Engineering · Vector Education

04/2023 — 04/2024

- Built and deployed backend services for Vector — an educational publishing app shipped to iOS and Android; onboarded 400 users.

Stack — Node.js, TypeScript, MongoDB, AWS

Software Engineer (Part-time) · WispSwap Lab

03/2023 — 01/2024

Remote

- Designed an efficient off-chain system for the WispSwap DeFi ecosystem; operated services for ~20,000 users; scaled to 1,000+ concurrent users in a single day.

Stack — TypeScript, Node.js, PostgreSQL, Redis

Software Engineer L3 · Teko

12/2021 — 06/2023

Enterprise CRM and merchant tooling for a major retail tech group.

- Built enterprise CRM apps on FastAPI, Elasticsearch, Redis, MySQL, and Kafka — cut API latency by 60% via targeted caching.
- Maintained legacy systems while upgrading the codebase with SOLID principles and design patterns; kept test coverage above 95%.
- Mentored four new team members through code reviews and on-the-job guidance.

Stack — Python, FastAPI, MySQL, Redis, Elasticsearch, Kafka

Software Engineer · Intern · Teko · VNIST

01/2020 — 11/2021

- Built two core microservices on FastAPI / PostgreSQL / Redis (500 users); integrated services to cut VNPAY merchant declaration time by 50%. Prior internship at VNIST.

Skills

Languages

TypeScript, Python, Go (learning), Solidity, Rust, C++, SQL

Backend & frameworks

Node.js, NestJS, FastAPI, microservices, REST, gRPC, WebSockets

Data & messaging

PostgreSQL, MySQL, Redis, Elasticsearch, MongoDB, Kafka, RabbitMQ

Cloud, CI/CD & observability

Docker, Kubernetes, GitHub Actions, AWS, OpenTelemetry, Prometheus / Grafana, structured logging

Web3

EVM, Solidity, TON / FunC, Sui, Solana, smart-contract design

Testing & practices

Unit + integration testing, TDD, code review, incident response, AI-assisted workflows, OOP & design patterns

Selected projects

ts-drp · Decentralized Realtime Program — TypeScript runtime for decentralized real-time collaboration — contributed CI/CD, test infrastructure (→ 90% coverage), and performance instrumentation. github.com/0xaldric/ts-drp

Mojave · Bitcoin / zk infrastructure — Rust workspace for full-node, sequencer, prover, and cryptographic utilities.

Contributed block signing & verification (PR #17) and automatic ethrex upgrade tooling (PR #20). github.com/1sixtech/mojave

AytuDex / TonPad / Pump.fun-style platform — Web3 systems: custom DEX routing (50–100% faster), TON launchpad smart contracts, scalable token issuance with separated read/write/socket layers.

FastAPI & Telegram Bot boilerplates — Open-source Python and NestJS starters — opinionated layouts, dependency injection, test-first structure.

Education

Hanoi University of Science and Technology (HUST) — B.Sc., Computer Science · 09/2017 — 09/2022

Achievements

- UAVS Hackatrix — Top 7, "Application for job management and salary payment" (07/2021)
- KPMG Challenge — solutions to help businesses overcome pandemic impact
- HackerRank — Problem-Solving Certificate